

Hotel Booking Exploratory Data Analysis (EDA)

Description

This project involves a comprehensive exploratory data analysis of the `hotel_booking.csv` dataset. The study analyzes 119,210 booking records across two distinct hotel types: City Hotels and Resort Hotels. Key findings indicate an overall cancellation rate of approximately 37.08%, with a strong seasonal peak in bookings during the summer months, particularly August and July.

Data Exploration

The dataset contains 36 features covering lead times, arrival dates, guest demographics, and booking channels.

- **Hotel Types:** City Hotels are the most popular, accounting for 66.41% of total bookings, while Resort Hotels comprise 33.59%.
- **Customer Composition:** "Transient" customers represent the largest segment at 75.06%.
- **Geographic Origin:** The majority of guests originate from Portugal (PRT), followed by Great Britain (GBR) and France (FRA).

Cancellations in Hotel Bookings

Similar to how weights determine the importance of information in deep learning, various factors in this dataset influence the "weight" or likelihood of a booking being canceled.

- **Overview:** Approximately 37% of all bookings were canceled.
- **Hotel-Specific Trends:** City Hotels experience a higher volume of cancellations compared to Resort Hotels, as shown in the visualization below.

Lead Time Analysis

Lead time represents the number of days between booking and arrival. In the context of model training and optimization, lead time can be seen as a predictive variable that helps the "system" (the hotel) recognize patterns in guest behavior.

- Average Lead Time: The mean lead time is 104.1 days.
- Pattern Recognition: The data shows a heavy skew toward shorter lead times, though long-term bookings (some over 700 days) exist and are more prone to cancellation.

Monthly Booking Distribution

The foundation of hospitality management relies on understanding spatial and temporal patterns. By learning separate trends for different months, hotels can optimize their resources.

- Peak Season: August (13,877 bookings) and July (12,661 bookings) are the busiest months.
- Off-Peak Season: January and February show the lowest booking counts.

Real-World Application: Revenue Management

Just as deep learning architectures like Xception are applied to skin disease classification in medical imaging, the insights from this hotel booking analysis are applied to dynamic pricing and overbooking strategies. By identifying that 37% of bookings cancel, revenue managers can "attach" new strategies (similar to classification heads) to optimize occupancy and prevent revenue loss from empty rooms.

Conclusion

This EDA serves as a "saved model" for hotel business logic. It captures the architecture of customer behavior and the "learned weights" of seasonality and lead times. These results are essential for reproducible business research and production deployment in automated booking systems.